

Dissecting neuronal substrates of positive and negative reinforcement in health and disease

My primary research goal is to separate the neuronal mechanisms for positive and negative reinforcement and determine how these processes are recruited and transformed throughout the development and expression of substance use disorder (SUD). My first projects will focus on the signaling and function of subpopulations in the lateral hypothalamus (LHA) and their regulation of dopamine (DA) signaling during positive, negative, and opioid reinforcement. My long-term goal is to identify interventions that can selectively modulate positive and negative reinforcement for the treatment of SUD.

Background and Vision The development of SUD is characterized in part by a transition from positive reinforcement to negative reinforcement¹, but the neuronal circuits that mediate these forms of reinforcement remain unclear. Multiple avenues of research suggest the LHA shape motivation for opioids as the LHA cells express a diversity of opioid related markers²⁻⁴ and plays a role in opioid withdrawal⁵⁻⁷. The LHA is deeply interconnected with the midbrain dopamine neurons that project throughout the brain⁸. Through engaging DA release at specific downstream sites, the LHA is positioned to modulate motivation for opioids through multiple mechanisms, but the causal role of the LHA and its regulation of the DA system in motivating opioid self-administration remains poorly understood.

My lab will address these questions using cutting-edge imaging techniques across positive and negative reinforcement or throughout the development and expression of SUD. My lab will determine the longitudinal encoding of individual neurons using 2-photon imaging with micropipettes to record hundreds of tracked neurons⁹ and determine the in-vivo coupling of the LHA and downstream DA signaling using multi-site fiber photometry during behavior. I will employ the head-fixed procedures I have developed to model positive and negative reinforcement¹⁰ along with head-fixed intravenous self-administration to model SUD^{11,12}. I will utilize computational approaches including general linear models and others to determine the neural encoding of reinforcement. I hypothesize that subsets of LHA neurons exhibit distinct encoding of positive and negative reinforcement that selectively regulates DA release in specific downstream areas to motivate drug taking.

Untangling the neuronal encoding of positive and negative reinforcement will inform therapeutic approaches that can selectively shape these mechanisms. Some of the projects outlined here are supported by a **NIDA K99/R00**.

Project 1: LHA and striatal DA encoding and function in positive and negative reinforcement. The LHA contains GABA and Glut neurons that drive opposing reinforcement phenotypes and regulation of dopamine in the ventral striatum^{3,13-15}, but the role of these populations during positive/negative reinforcement remains poorly understood. I will determine the dynamics of tracked neurons using 2-photon imaging and the relation to downstream DA signaling using multi-site fiber photometry. I will then test the causal role of this circuit using chemogenetics and disconnect approaches. This project will establish a framework for how the LHA regulates the DA system to mediate positive/negative reinforcement that may be recruited during the development of SUD.

Project 2: LHA and striatal DA encoding during development and expression of motivation for opioids. The role of DA in SUD is well established but the spatiotemporal DA dynamics during opioid self-administration and the role of descending LHA regulation of the DA system in SUD remains poorly understood. I will utilize approaches outlined in Project 1 to determine how this circuit regulates motivation for opioids.

Project 3: Linking activity-defined clusters of LHA neurons to circuit and behavior. It is unknown if there are discrete clusters of neurons in the LHA that selectively modulate subcomponents of behavior and DA release in subregions of the striatum. I will use 2-photon imaging during reinforcement behavior to identify clusters of cells that show unique activity dynamics and then measure the behavioral function of these cells using holographic stimulation⁹ during behavior. I will also assess the relationship between these cells and downstream DA by multiplexing 2-photon imaging and multi-site fiber photometry.

Past Research Accomplishments: Postdoctoral Research: I developed a novel head-fixed system for measuring a range of consumption and reinforcement behaviors in head-fixed mice¹⁰, and used this system to assess the role the LHA and DA release in consumption (*in preparation*). I conducted an exhaustive review of the signaling and function of subcortical GABA and Glutamate in reinforcement / valence¹⁶. I have also shared my technical expertise in designing and implementing head-fixed approaches as a co-author a primer in *Neuron*¹⁷ and a protocols paper on head-fixed IVSA (*in preparation*)

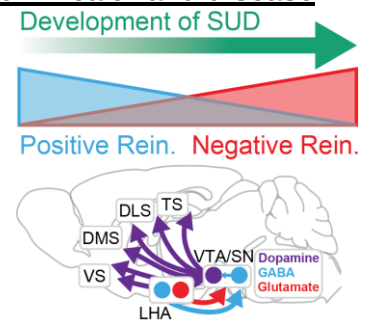


Figure 1: Research Question.

How does the LHA regulate striatal DA to motivate positive negative reinforcement throughout the development of SUD?

References

1. Koob, G. F. & Moal, M. L. Drug Abuse: Hedonic Homeostatic Dysregulation. *Science* **278**, 52–58 (1997).
2. Mansour, A., Khachaturian, H., Lewis, M. E., Akil, H. & Watson, S. J. Anatomy of CNS opioid receptors. *Trends Neurosci* **11**, 308–314 (1988).
3. Rossi, M. A. *et al.* Obesity remodels activity and transcriptional state of a lateral hypothalamic brake on feeding. *Science* **364**, 1271–1274 (2019).
4. Mansour, A., Fox, C. A., Akil, H. & Watson, S. J. Opioid-receptor mRNA expression in the rat CNS: anatomical and functional implications. *Trends Neurosci* **18**, 22–29 (1995).
5. Sheng, H. *et al.* Nucleus accumbens circuit disinhibits lateral hypothalamus glutamatergic neurons contributing to morphine withdrawal memory in male mice. *Nat Commun* **14**, 71 (2023).
6. Zhou, Y. *et al.* Mu opioid receptor and orexin/hypocretin mRNA levels in the lateral hypothalamus and striatum are enhanced by morphine withdrawal. *J Endocrinol* **191**, 137–145 (2006).
7. Georgescu, D. *et al.* Involvement of the Lateral Hypothalamic Peptide Orexin in Morphine Dependence and Withdrawal. *J Neurosci* **23**, 3106–3111 (2003).
8. Menegas, W. *et al.* Dopamine neurons projecting to the posterior striatum form an anatomically distinct subclass. *eLife* **4**, e10032 (2015).
9. Hjort, M. M. *et al.* Microprisms enable enhanced throughput and resolution for longitudinal tracking of neuronal ensembles in deep brain structures. *Neurophotonics* **11**, (2024).
10. Gordon-Fennell, A. *et al.* An open-source platform for head-fixed operant and consummatory behavior. *eLife* **12**, e86183 (2023).
11. Vollmer, K. M. *et al.* A Novel Assay Allowing Drug Self-Administration, Extinction, and Reinstatement Testing in Head-Restrained Mice. *Front Behav Neurosci* **15**, 744715 (2021).
12. Vollmer, K. M. *et al.* An opioid-gated thalamoaccumbal circuit for the suppression of reward seeking in mice. *Nat Commun* **13**, 6865 (2022).
13. Jennings, J. H. *et al.* Visualizing Hypothalamic Network Dynamics for Appetitive and Consummatory Behaviors. *Cell* **160**, 516–27 (2015).
14. Jennings, J. H., Rizzi, G., Stamatakis, A. M., Ung, R. L. & Stuber, G. D. The Inhibitory Circuit Architecture of the Lateral Hypothalamus Orchestrates Feeding. *Science* **341**, 1517–1521 (2013).
15. Nieh, E. H. *et al.* Decoding neural circuits that control compulsive sucrose seeking. *Cell* **160**, 528–41 (2015).
16. Gordon-Fennell, A. & Stuber, G. D. Illuminating subcortical GABAergic and glutamatergic circuits for reward and aversion. *Neuropharmacology* **198**, 108725 (2021).
17. Zhou, Z. C. *et al.* Deep-brain optical recording of neural dynamics during behavior. *Neuron* **111**, 3716–3738 (2023).